

2012 Spring MAE 143B HW4

Due date: 05/03

- ✓4.22 A linear ODE model of the DC motor with negligible armature inductance ($L_a = 0$) and with a disturbance torque w was given earlier in the chapter; it is restated here, in slightly different form, as

$$\frac{JR_a}{K_t} \ddot{\theta}_m + K_e \dot{\theta}_m = v_a + \frac{R_a}{K_t} w,$$

where θ_m is measured in radians. Dividing through by the coefficient of $\ddot{\theta}_m$, we obtain

$$\ddot{\theta}_m + a_1 \dot{\theta}_m = b_0 v_a + c_0 w,$$

where

$$a_1 = \frac{K_t K_e}{JR_a}, \quad b_0 = \frac{K_t}{JR_a}, \quad c_0 = \frac{1}{J}.$$

With rotating potentiometers, it is possible to measure the positioning error between θ and the reference angle θ_r or $e = \theta_{ref} - \theta_m$. With a tachometer we can measure the motor speed $\dot{\theta}_m$. Consider using feedback of the error e and the motor speed $\dot{\theta}_m$ in the form

$$v_a = K(e - T_D \dot{\theta}_m),$$

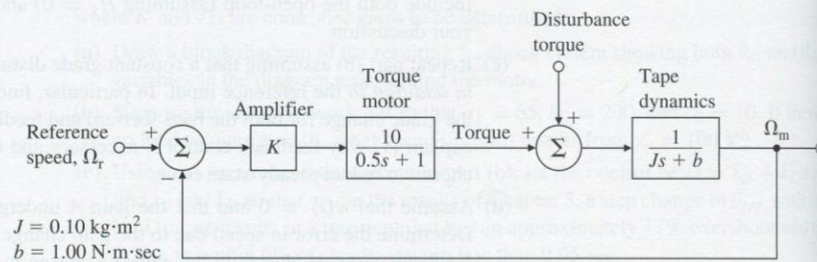
where K and T_D are controller gains to be determined.

- (a) Draw a block diagram of the resulting feedback system showing both θ_m and $\dot{\theta}_m$ as variables in the diagram representing the motor.
- (b) Suppose the numbers work out so that $a_1 = 65$, $b_0 = 200$, and $c_0 = 10$. If there is no load torque ($w = 0$), what speed (in rpm) results from $v_a = 100$ V?
- (c) Using the parameter values given in part (b), let the control be $D = k_p + k_D s$ and find k_p and k_D so that, using the results of Chapter 3, a step change in θ_{ref} with zero load torque results in a transient that has an approximately 17% overshoot and that settles to within 5% of steady-state in less than 0.05 sec.
- (d) Derive an expression for the steady-state error to a reference angle input, and compute its value for your design in part (c) assuming $\theta_{ref} = 1$ rad.
- (e) Derive an expression for the steady-state error to a constant disturbance torque when $\theta_{ref} = 0$ and compute its value for your design in part (c) assuming $w = 1.0$.

- 4.26 The transfer functions of speed control for a magnetic tape-drive system are shown in Fig. 4.39. The speed sensor is fast enough that its dynamics can be neglected and the diagram shows the equivalent unity feedback system.
- Assuming the reference is zero, what is the steady-state error due to a step disturbance torque of $1 \text{ N} \cdot \text{m}$? What must the amplifier gain K be in order to make the steady-state error $e_{ss} \leq 0.01 \text{ rad/sec}$?
 - Plot the roots of the closed-loop system in the complex plane, and accurately sketch the time response of the output for a step reference input using the gain K computed in part (a).
 - Plot the region in the complex plane of acceptable closed-loop poles corresponding to the specifications of a 1% settling time of $t_s \leq 0.1 \text{ sec}$ and an overshoot $M_p \leq 5\%$.
 - Give values for k_p and k_D for a PD controller, which will meet the specifications.
 - How would the disturbance-induced steady-state error change with the new control scheme in part (d)? How could the steady-state error to a disturbance torque be eliminated entirely?

Figure 4.39

Speed-control system for a magnetic tape-drive



- 4.29 The DC-motor speed control shown in Fig. 4.41 is described by the differential equation

$$\dot{y} + 60y = 600v_a - 1500w,$$

where y is the motor speed, v_a is the armature voltage, and w is the load torque. Assume the armature voltage is computed using the PI control law

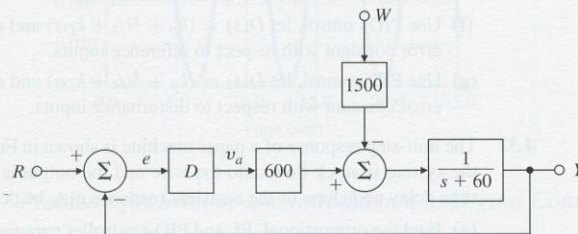
$$v_a = -\left(k_p e + k_I \int_0^t e dt\right),$$

where $e = r - y$.

- Compute the transfer function from W to Y as a function of k_p and k_I .
- Compute values for k_p and k_I so that the characteristic equation of the closed-loop system will have roots at $-60 \pm 60j$.

Figure 4.41

DC Motor speed-control block diagram for Problems 4.29 and 4.30



- 4.30 For the system in Problem 4.29, compute the following steady-state errors:

- to a unit-step reference input;
 - to a unit-ramp reference input;
 - to a unit-step disturbance input;
 - for a unit-ramp disturbance input.
- (e) Verify your answers to (a) and (d) using MATLAB. Note that a ramp response can be generated as a step response of a system modified by an added integrator at the reference input.